COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures an Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	07 / 23 /2026	**DATE SUBMITTED:**	
NAME:	Elijah I. DIzor	**INSTRUCTOR:**	Engr. Jimlord Quejado

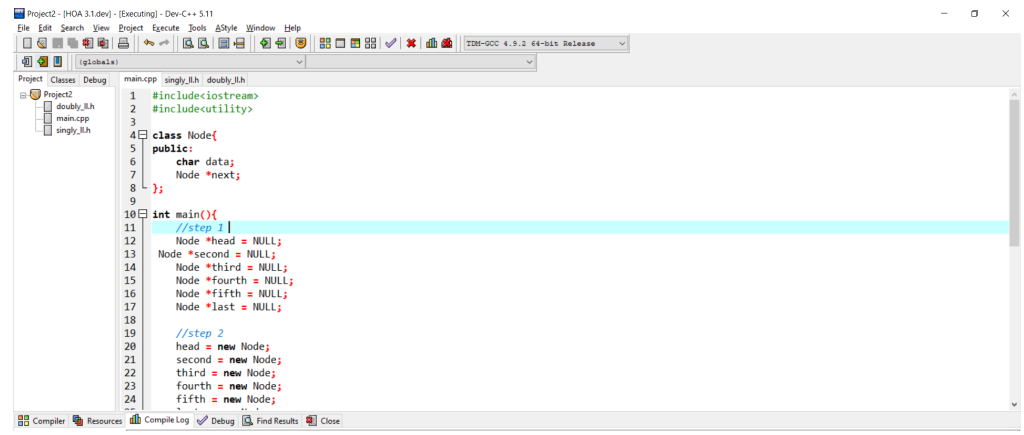
ACTIVITY NO. 3.1 [Linked List]

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Construct C++ code for a singly and doubly linked list in C++
A.1. Singly Linked List

CODE

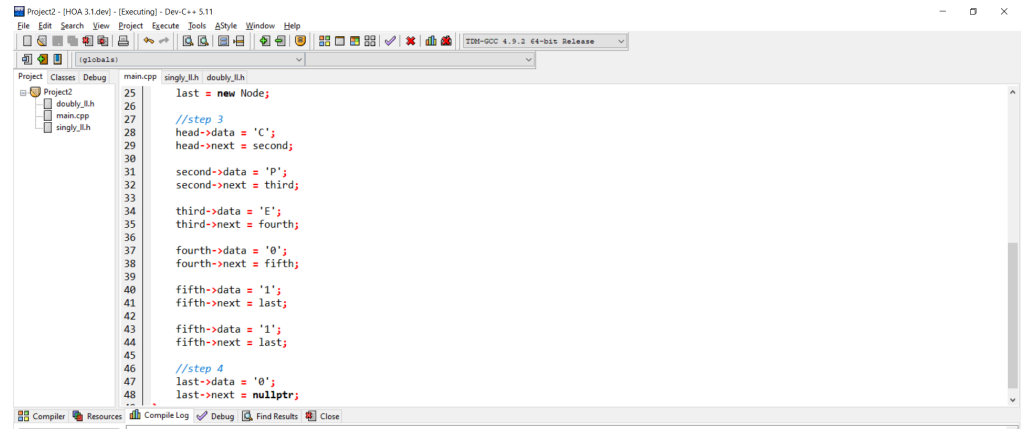


```
#include<iostream>
#include<utility>

class Node{
public:
    char data;
    Node *next;
};

int main(){
    //step 1
    Node *head = NULL;
 Node *second = NULL;
    Node *third = NULL;
    Node *fourth = NULL;
    Node *fifth = NULL;
    Node *last = NULL;

    //step 2
    head = new Node;
    second = new Node;
    third = new Node;
    fourth = new Node;
    fifth = new Node;
```



```
    last = new Node;

    //step 3
    head->data = 'C';
    head->next = second;

    second->data = 'P';
    second->next = third;

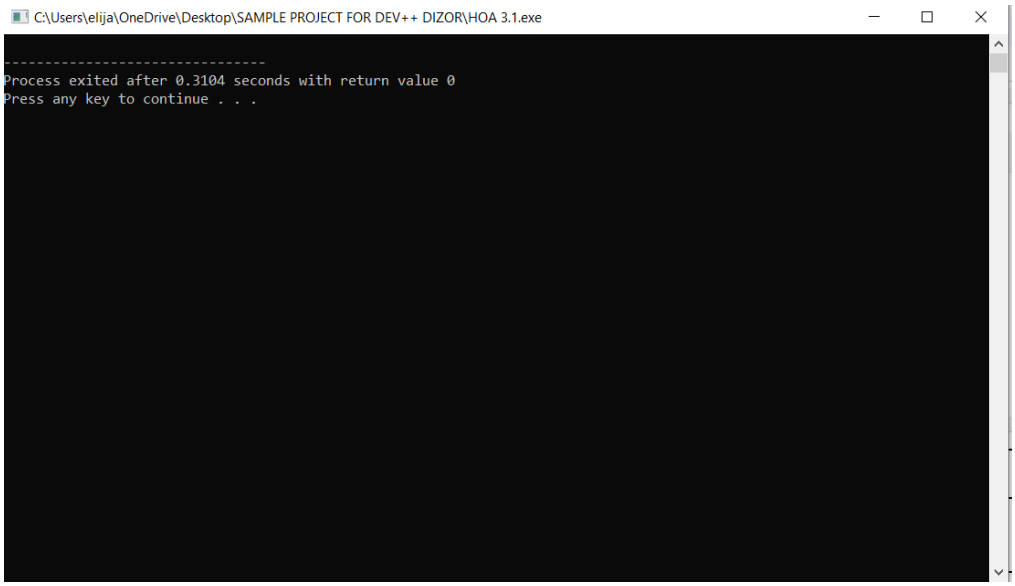
    third->data = 'E';
    third->next = fourth;

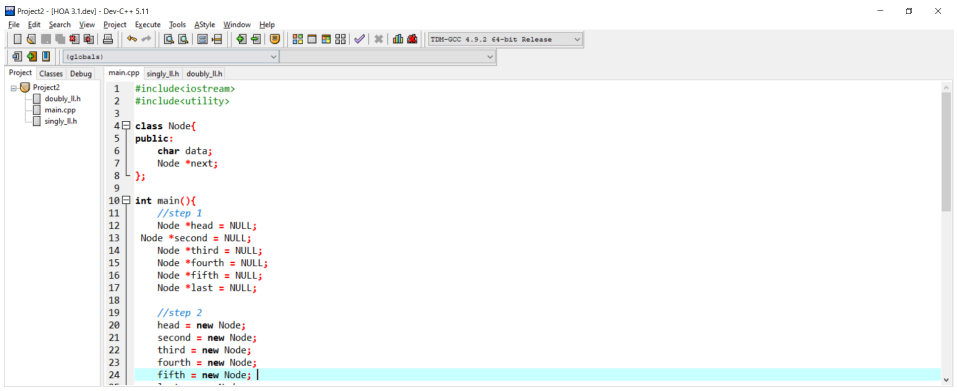
    fourth->data = '0';
    fourth->next = fifth;

    fifth->data = '1';
    fifth->next = last;

    fifth->data = '1';
    fifth->next = last;

    //step 4
    last->data = '0';
    last->next = nullptr;
```

OUTPUT	
OBSERVATION	The program works, but the output is blank because there are no std::cout statements. It can be improved by adding traversal logic to see the actual output CPE 010, clean up memory to prevent leaks, and transition.

Code	

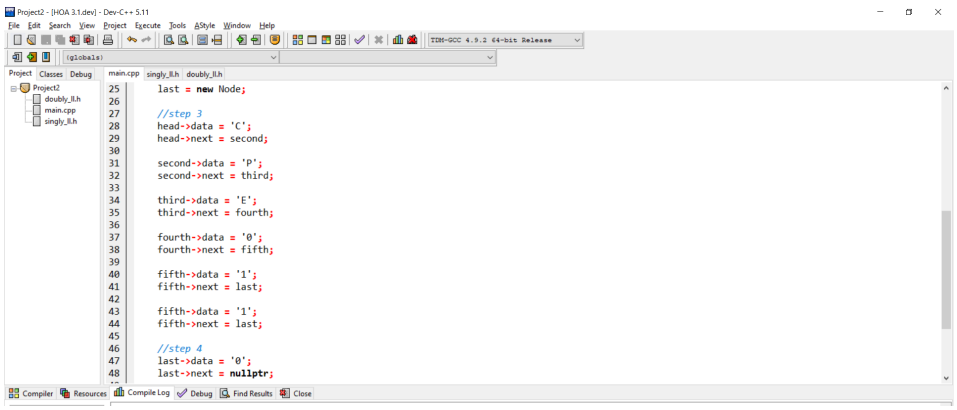
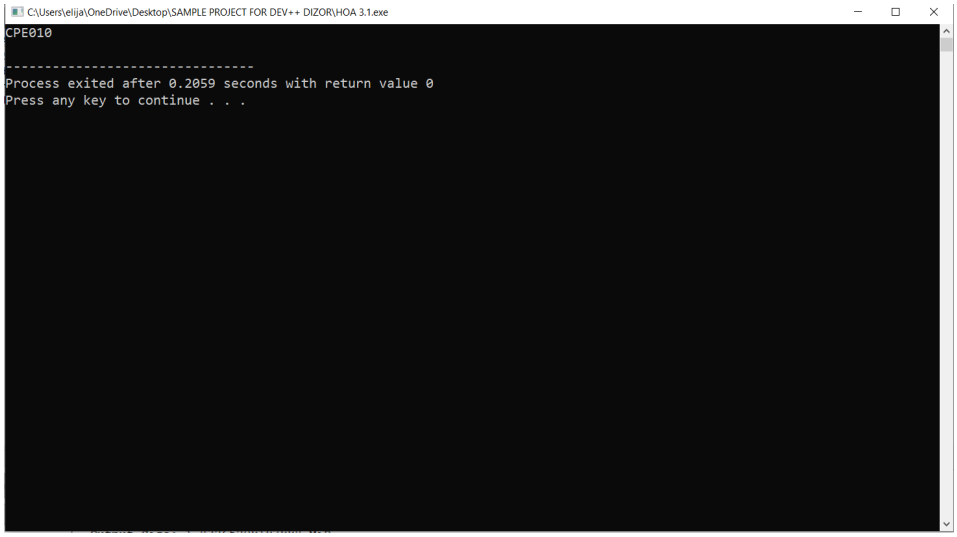
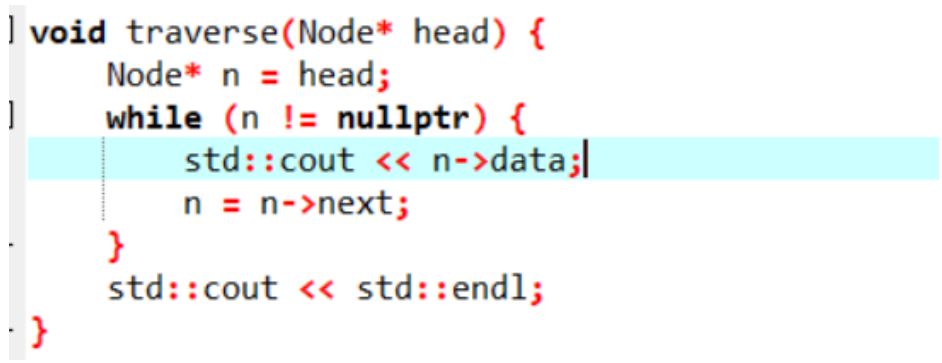
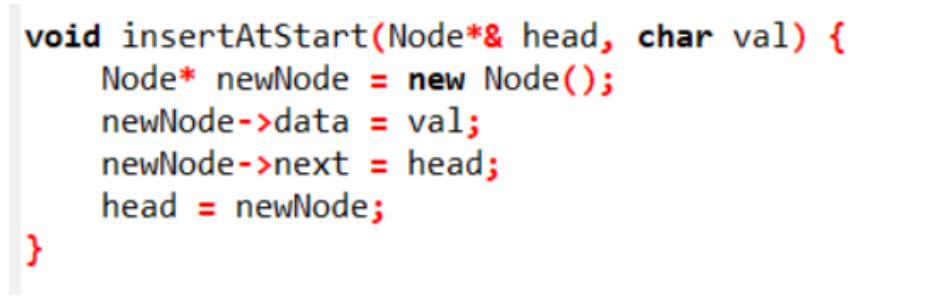
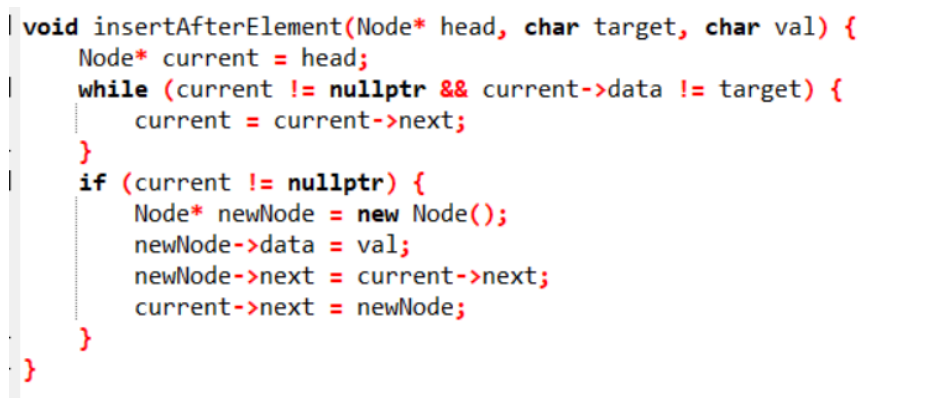
	
Screenshot	
Discussion	The program was able to show the output CPE 010. The structural links created dynamically on the heap are aligned with sequential logic. The loop ends when n is updated to last->next, because it evaluates strictly to nullptr, the condition n ! = nullptr becomes false.

Table 3-1. Output of initial/Simple Implementation

Operation	Screenshot

Traversal	
Insertion at head	
Insertion at any part of the list	

Insertion at the end	`void insertAtEnd(Node* head, char val) {`
	`  Node* newNode = new Node();`
	`  newNode->data = val;`
	`  newNode->next = nullptr;`
	`  if (head == nullptr) return;`
	`  Node* temp = head;`
	`  while (temp->next != nullptr) {`
	`  temp = temp->next;`
	`  }`
	`  temp->next = newNode;`
	`}`

Depletion of a node	`void deleteNode(Node*& head, char target) {`
	`  if (head == nullptr) return;`
	
	`  if (head->data == target) {`
	`  Node* temp = head;`
	`  head = head->next;`
	`  delete temp;`
	`  return;`
	`  }`

Table 3-2. Code for the list operations

a	Source code	`std::cout << "a. initial list: ";` `traverse(head);`
	Console	`a. initial list: CPE101`

b	Source code	`insertAtStart(head, 'G');`
		`std::cout << "b. after inserting 'G' at start: ";`
		`traverse(head);`

	Console	`b. after inserting 'G' at start: GCPE101`

c	Source Code	`insertAfterElement(head, 'P', 'E');`
		`std::cout << "c. after inserting 'E' after 'P': ";`
		`traverse(head);`

	Console	`c. after inserting 'E' after 'P': GCPEE101`

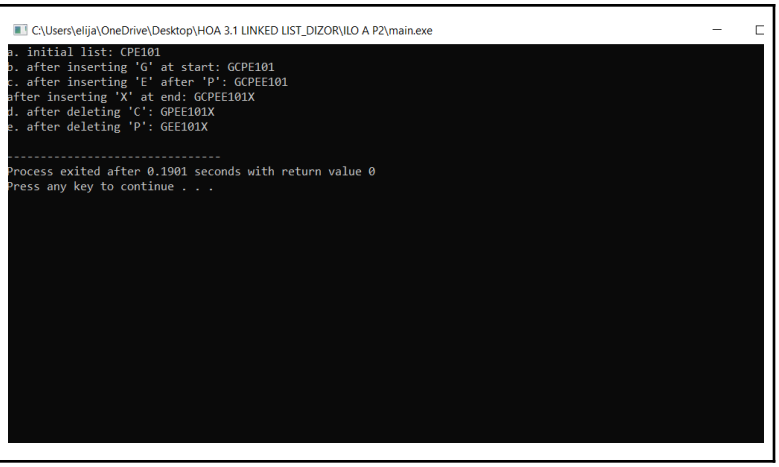
d	Source Code	`deleteNode(head, 'C');`
		`std::cout << "d. after deleting 'C': ";`
		`traverse(head);`

	Console	`d. after deleting 'C': GPEE101X`

e	Source Code	`deleteNode(head, 'P');`
		`std::cout << "e. after deleting 'P': ";`
		`traverse(head);`

	Console	`e. after deleting 'P': GEE101X`

f	Source Code	`traverse(head);`

	Console	

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO B: Solve given problems utilizing linked lists in C++ Problem Title: Implementing a Song Playlist using Linked List Source: Packt Publishing

Problem Description: In this activity, we'll look at some applications for which a singly linked list is not enough or not convenient. We will build a tweaked version that fits the application. We often encounter cases where we have to customize default implementations, such as when looping songs in a music player or in games where multiple players take a turn one by one in a circle.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **chatgpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of concepts and code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ai to explain concepts and to understand how singly and doubly affects the output of main.cpp, and how linked lists are made.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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